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PHYSICS PTA - MODEL QUESTION PAPER -3
STANDARD XII

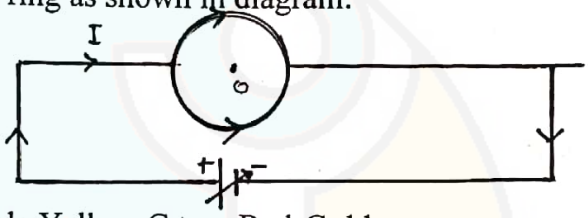
TIME ALLOWED : 15min + 3.00 Hr **Maximum Marks : 70**

PART – I

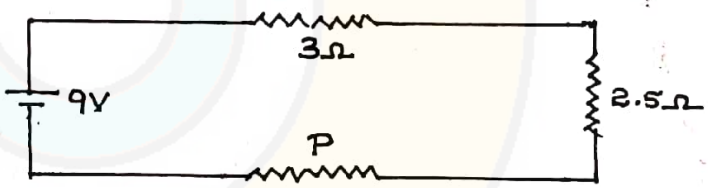
Note: (i) Answer all the questions **15× 1 =15**

(ii) Choose the most appropriate answer from four given alternatives and write the option code with the corresponding answer

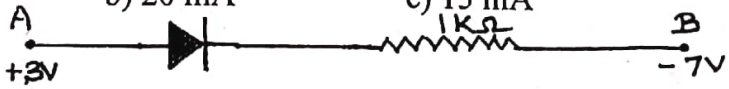
1. Cellular phones use radio waves in ____ band
   a) long wave b) short wave c) medium wave **d) ultra high frequency**
2. Dimension and unit of Electric flux is _____
   a) $ML^2T^{-3}A^{-2}, Nm^2C^{-1}$ **b) $ML^3T^{-3}A^{-1}, Nm^2C^{-1}$**
   c) $ML^2T^{-1}A^{-2}, Nm^2C^{-1}$ d) $ML^{-4}T^{-3}A^{-2}, Nm^2C^{-1}$
3. RMS voltage and frequency of V= 230 sin(314t) A.C. source.
   a) 162.6V, 50Hz b) 230V, 50Hz c) 230V, 60Hz d) 162.6 V, 25Hz
4. Magnetic induction at the centre of the ring as shown in diagram.



   a) 0.2 T b) 0.5 T
   c) 0 T d) 1.0 T
5. Resistance of resistor having colour code Yellow-Green-Red-Gold
   a) (4700±5%)Ω **b) (4500±5%)Ω** c) (4700±10%)Ω d) (4500±10%)Ω
6. There is a current of 1A in the circuit shown below. What is the resistance of P?



   a) 1.5Ω b) 2.5Ω
   c) 3.5Ω d) 4.5Ω
7. Three wires of equal lengths are bent in the form of loops. One of the loops is circle, another is a semi-circle and the third one is a square. They are placed in a uniform magnetic field and same electric current is passed through them. Which of the following loop configuration will experience greater torque?
   a) circle b) semi – circle c) square d) all of them
8. Rainbow is formed by _____ of light by droplets of water.
   a) dispersion b) partial polarisation c) plane of polarization d) interference
9. 1 eV is equal to
   a) 1.602×10^{19}J **b) 1.602×10^{-19}J** c) 1.602×10^{-29}J d) 1.602×10^{29}J
10. light transmitted by Nicol prism is,
    a) partially polarized b) unpolarised **c) plane polarized** d) elliptically polarized
11. The radius of the 5th orbit of hydrogen atom is 13.25Å. Calculate the wavelength of the electron in the 5th orbit.
    a) λ = 16.64Å b) λ = 26.50Å c) λ = 13.25Å d) λ = 13.64Å
12. Half life of an element is 25 years, then what percentage of the sample will be left over after 125 years and its life time are
    a) 3.125% & 1250 **b) 3.125% & ∞** c) 6.25% & 1250 d) 6.25 & ∞
13. Consider an ideal junction diode. Find the value of current flowing through AB is
    a) 10 mA b) 20 mA c) 15 mA d) 11 mA



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14. The frequency range of 3MHz to 30 MHz is used for
a) ground wave propagation b) space wave propagation
c) sky wave propagation d) satellite communication

15. The strongest source of gravitational waves is
a) black holes b) accelerated mass c) god particles d) all the above

PART – II

Answer any six questions . Question number 24 compulsory. $6 \times 2 = 12$

16. State Fleming's right hand rule.

The thumb, index finger and middle finger of right hand are stretched out in mutually perpendicular directions. If the index finger points the direction of the magnetic field and the thumb indicates the direction of motion of the conductor, then the middle finger will indicate the direction of the induced current.

17. Why do clouds appear white?

If the light pass through the clouds which contains large amount of dust and water droplets, all the colours get equally scattered irrespective of wavelength. This is the reason for the whitish appearance of cloud.

18. Define the physical quantity whose unit is Vm, and state whether it is scalar or vector.

Vm is the unit of electric flux. It is a scalar.

The number of electric field lines crossing a given area kept normal to the electric field lies is called electric flux

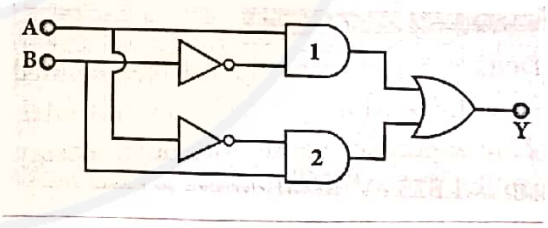
19. State de Broglie hypothesis

According to de Broglie hypothesis, all matter particles like electrons, protons, neutrons in motion are associated with waves. These waves are called de Broglie waves or matter waves.

20. What does RADAR stand for?

Radar basically stands for Radio Detection and Ranging System.

21. In a combination of the following gates, write the Boolean equation for output Y in terms of inputs A and B.



Output of the first AND gate = $A \cdot \overline{B}$

Output of the second AND gate = $\overline{A} \cdot B$

Output of the OR gate = $Y = A \cdot \overline{B} + \overline{A} \cdot B$

22. Calculate the radius of $^{197}_{79}Au$ nucleus

$R = R_0 A^{\frac{1}{3}}$

$= 1.2 \times 10^{-15} \times (197)^{\frac{1}{3}}$

$= 1.2 \times 10^{-15} \times 5.818$

$= 6.9816 \times 10^{-15}$

$= 6.9816$ F

23. Mention any two advantages and disadvantages of Robotics.

Advantages of Robots: (any two)

i) They are much cheaper than humans

ii) They never get tired like humans. It can work 24 × 7. Hence absenteeism in work place can be reduced

iii) They are more precius and error free in performing the task

iv) They are stronger and faster than humans

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Disadvantages of Robots

(i) Robats have no sense of emotions or conscience
(ii) They lack empathy and hence create an emotionless workplace
(iii) Unemployment problem will increase.
(iv) Humans cannot be replaced by robots in decision making

24. A circular coil of radius 5cm and 50 turns carries a current of 3 ampere. Find the magnetic dipole moment of the coil.

Magnetic dipole moment $\vec{p}_m = I\vec{A}$

For n turns $\vec{p}_{m(net)} = n\,I\vec{A}$

$= 50 \times 3 \times 3.14 \times (5 \times 10^{-2})^2$

$= 11775 \times 10^{-4}\ Am^2 = 1.1775\ Am^2 \approx 1.18\ Am^2$

PART - III

Answer any six questions . Question number 33 compulsory. $6 \times 3 = 18$

25. Derive an expression for the torque experienced by a dipole due to uniform electric field.

Consider an electric dipole of dipole moment placed in a uniform electric field whose field lines are equally spaced and point in the same direction.

The charge +q will experience a force $q\vec{E}$ in the direction of the field and charge –q will experience a force $-q\vec{E}$ in a direction opposite to the field. Since the external field is uniform, the total force acting on the dipole is zero.

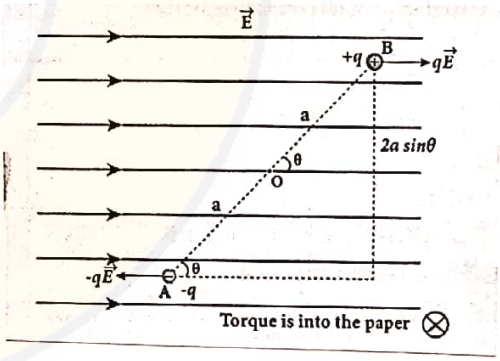
These two forces acting at different points will constitute a couple and the dipole experience a torque as shown in Figure.

This torque tends to rotate the dipole.

The total torque on the dipole about the point O

$\vec{\tau} = \overrightarrow{OA} \times (-qE) + \overrightarrow{OB} \times qE$ .....(1)

Using right-hand corkscrew rule,
it is found that total torque is perpendicular
to the plane of the paper and is directed into it.



The magnitude of the total torque

$\tau = |\overrightarrow{OA}| \times |(-qE)| \sin\theta + |\overrightarrow{OB}| \times |qE| \sin\theta$ ...(2)

$\tau = qE\ 2a \sin\theta$

$\tau = pE \sin\theta$

where θ is the angle made by $\vec{p}$ with $\vec{E}$. Since p = 2aq,

The torque is written in terms of the vector product as $\vec{\tau} = \vec{p} \times \vec{E}$

26. Find the heat energy produced in a resistance of 10Ω when 5A current flows through it for 5 minutes.

$H = I^2Rt$

$= 5^2 \times 10 \times 300$

$= 75000\ J$

27. Show that the magnetic field at any point on the axis of the solenoid having n turns per unit length is $B = \frac{1}{2}\mu_0 nI(\cos\theta_1 - \cos\theta_2)$

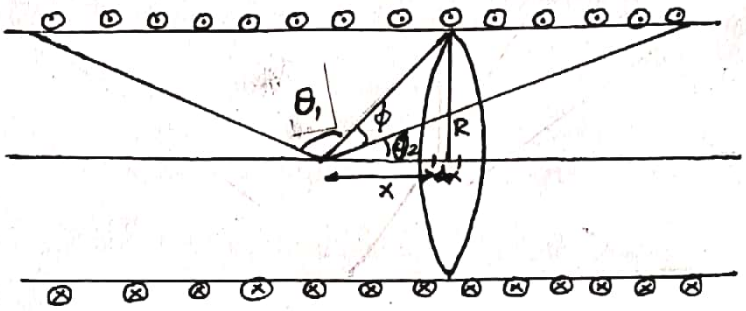
Number of turns per unit length = n

For length dx, total number of turns = n dx

$dB = \frac{\mu_0 I R^2 n dx}{2(R^2+x^2)^{\frac{3}{2}}}$

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From the figure,

$\frac{R}{x} = \text{Tan}\phi$; $\frac{x}{R} = \cot\phi$

$x = R\cot\phi$; $dx = -R\,\text{cosec}^2\phi\, d\phi$

$$dB = \frac{\mu_o I R^2 n(-R\,\text{cosec}^2\phi\, d\phi)}{2(R^2+R^2\cot^2\phi)^{\frac{3}{2}}}$$

$$= -\frac{\mu_o I R^3 n\,\text{cosec}^2\phi\, d\phi}{2R^3(1+\cot^2\phi)^{\frac{3}{2}}}$$

$$= -\frac{\mu_o I n\,\text{cosec}^2\phi\, d\phi}{2\text{cosec}^3\phi}$$

$$= -\frac{\mu_o I n\, d\phi}{2\text{cosec}\,\phi}$$

$\because\ (1+\cot^2\phi)^{\frac{3}{2}} = (\text{cosec}^2\phi)^{\frac{3}{2}} = \text{cosec}^3\phi$

$$B = \int_{\theta_2}^{\theta_1} dB = -\frac{\mu_o I n}{2}\int_{\theta_2}^{\theta_1} \sin\phi\, d\phi$$

$$= -\frac{\mu_o I n}{2}[-\cos\phi]_{\theta_2}^{\theta_1} = -\frac{\mu_o I n}{2}[-\cos\theta_1 + \cos\theta_2\ ]$$

$$B = \frac{1}{2}\mu_0 n I(\cos\theta_1 - \cos\theta_2)$$

28. State and obtain Malus' law.

Malus discovered that when a beam of plane polarised light of intensity I_0 is incident on an analyser, the light transmitted of intensity I from the analyser varies directly as the square of the cosine of the angle θ between the transmission axis of polariser and analyser . This is known as Malus' law.

$$I = I_0\cos^2\theta$$

Proof :

Let us consider the plane of polariser and analyser are inclined to each other at an angle θ is as shown in Figure .

Let I_0 be the intensity and *a* be the amplitude of the electric vector transmitted by the polariser. The amplitude *a* of the incident light has two rectangular components, (acosθ) parallel component and (asinθ) perpendicular component to the axis of transmission of the analyser.

Only the component (acosθ) will be transmitted by the analyser.

The intensity of light transmitted from the analyser is proportional to the square of the component of the amplitude transmitted by the analyser.

$I \propto (a\cos\theta)^2$

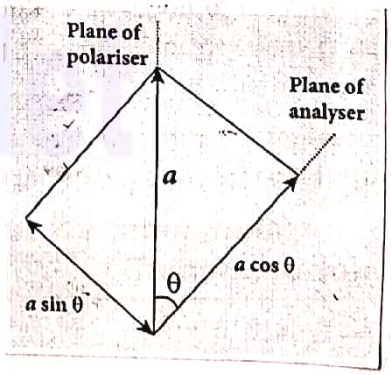
$I = k(a\cos\theta)^2$

Where k is constant of proportionality.

$I = ka^2\cos^2\theta$

$I = I_0\cos^2\theta$

Where, $I_o = ka^2$ is the maximum intensity of light transmitted from the analyser.



29. Show that Lenz's law is in accordance with the law of conservation of energy.

According to Lenz's law, when a magnet is moved either towards or away from a coil, the induced current produced opposes its motion.

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As a result, there will always be a resisting force on the moving magnet. So work has to be done by some external agency to move the magnet against this resistive force.

Here the mechanical energy of the moving magnet is converted into the electrical energy which in turn gets converted in to Joule heat in the coil. (i.e) energy is conserved from one form to another

On the contrary to Lenz's law, let us assume that the induced current helps the cause responsible for its production.

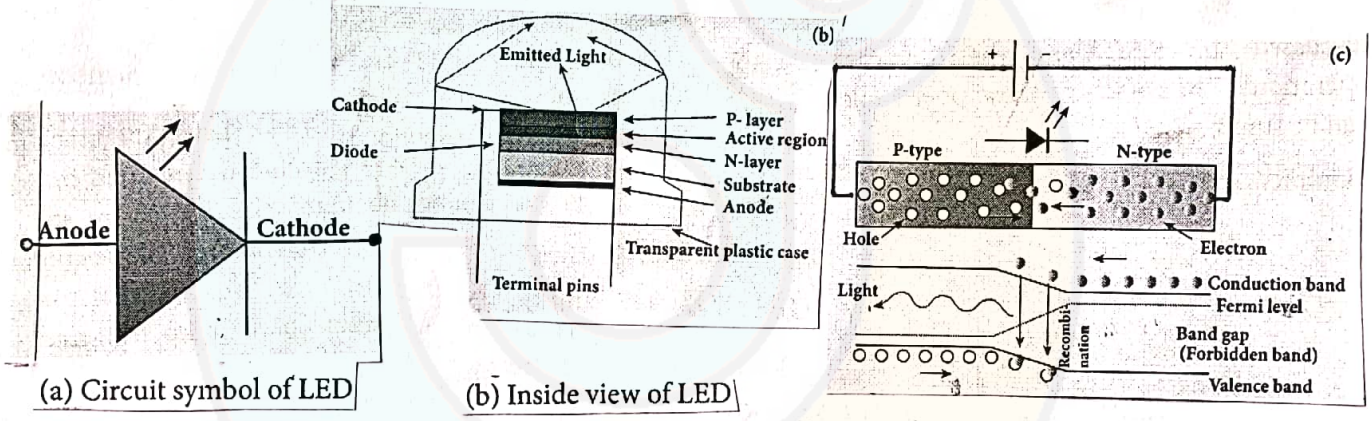
If we push the magnet little bit towards the coil, the induced current helps the movement of the magnet towards the coil.

Then the magnet starts moving towards the coil without any expense of energy. This, then, becomes a perpetual motion machine.

In practice, no such machine is possible. Therefore, the assumption that the induced current helps the cause is wrong.

30. What is an LED? Give the principle of operation with the diagram.

LED is a p-n junction diode which emits visible or invisible light when it is forward biased. Since, electrical energy is converted into light energy, this process is also called electro luminescence.



The circuit symbol of LED is shown in Figure (a).

The cross-sectional view of a commercial LED is shown in Figure(b)

It consists of a p-layer, n-layer and a substrate.

A transparent window is used to allow light to travel in the desired direction. An external resistance in series with the biasing source is required to limit the forward current through the LED.

In addition, it has two leads; anode and cathode. When the p-n junction is forward biased, the conduction band electrons on n-side and valence band holes on p-side diffuse across the junction.

When they cross the junction, they become excess minority carriers (electrons in p-side and holes in n-side).

These excess minority carriers recombine with oppositely charged majority carriers in the respective regions, i.e. the electrons in the conduction band recombine with holes in the valence band as shown in the Figure (c).

During recombination process, energy is released in the form of light (radiative) or heat (non-radiative).

For radiative recombination, a photon of energy hv is emitted. For non-radiative recombination, energy is liberated in the form of heat.

The colour of the light is determined by the energy band gap of the material.

Therefore, LEDs are available in a wide range of colours such as blue (SiC), green (AlGaP) and red (GaAsP). Now a days, LED which emits white light (GaInN) is also available.

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31. Briefly explain the elementary particles of nature.

An atom has a nucleus surrounded by electrons and nuclei is made up of protons and neutrons. Till 1960s, it was thought that protons, neutrons and electrons are fundamental building blocks of matter.

In1964, physicist Murray Gellman and GeorgeZweig theoretically proposed that protons and neutrons are not fundamental particles , in fact they are made up of quarks.

These quarks are now considered elementary particles of nature.

Electrons are fundamental or elementary particles because they are not made up of anything.

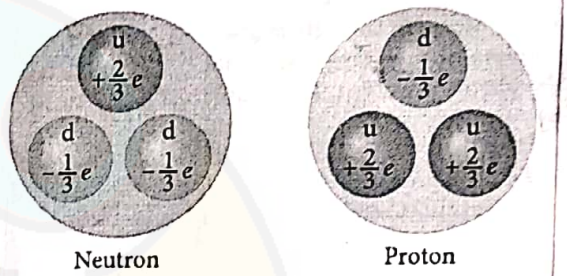
In the year 1968, the quarks were discovered experimentally by Stanford Linear Accelerator Center (SLAC), USA.

There are six quarks namely, **up, down, charm, strange, top and bottom** and their antiparticles. All these quarks have fractional charges.

For example, charge of up quark is $+\frac{2}{3}e$

and that of down quark is $-\frac{1}{3}e$

According to quark model,
proton is made up of two up quarks and one down quark and
neutron is made up of one up quark and two down quarks as shown in the figure



Constituents of nucleons

32. Explain the Maxwell's modification of Ampere's circuital law.

According to Faraday's law of electromagnetic induction which states that the change in magnetic field produces an electric field.

The magnetic field $\vec{B}$ induced along a closed loop by the changing electric flux Φ_E in the region encircled by that loop, this is known as Maxwell's law of induction.

In order to understand how the changing electric field produces magnetic field, let us consider a situation of 'charging a parallel plate capacitor' shown in Figure (1)

The electric current passing through the wire is the conduction current Ic. This current generates magnetic field around the wire.

According to Ampere's circuital law $\oint_l \vec{B}.\vec{dl} = \mu_0 I(t)$ -----(1)

where μ_0 permeability of free space

To calculate the magnetic field at a point P near the wire as shown in Figure(2) let us draw an amperian loop (circular loop) which encloses the surface S_1 (circular surface). Therefore, using Ampere's circuital law , we get

$$\oint_l \vec{B}.\vec{dl} = \mu_0 I_C \text{ -----(2)}$$

where I_C is the conduction current.

Suppose the same loop is enclosed by balloon shaped surface S_2 as shown in Figure (3)

$$\oint_{\text{enclosing } s_2} \vec{B}.\vec{dl} = 0\text{-----(3)}$$

Due to external source (battery or cell), the capacitor gets charged up because of current flowing through the capacitor. This produces an increasing electric field between the capacitor plates.

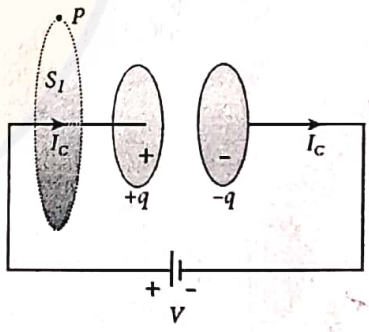


Figure 5.4 Applying Ampere's circuital law - loop enclosing surface

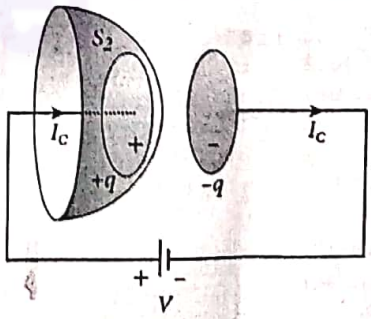


Figure 5.5 Applying Ampere's circuital law - loop enclosing surface S2

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From Gauss's law the electric flux between the plates of the capacitor is

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = EA = \frac{q}{\varepsilon_o} \quad \text{----(4)}$$

Where A is the area of the plate of capacitor.

The change in electric flux is

$$\frac{d\phi_E}{dt} = \frac{1}{\varepsilon_o}\frac{dq}{dt}$$

$$\frac{dq}{dt} = \varepsilon_o \frac{d\phi_E}{dt} \quad \text{-----(5)}$$

$$\frac{dq}{dt} = I_d \text{----------(6)}$$

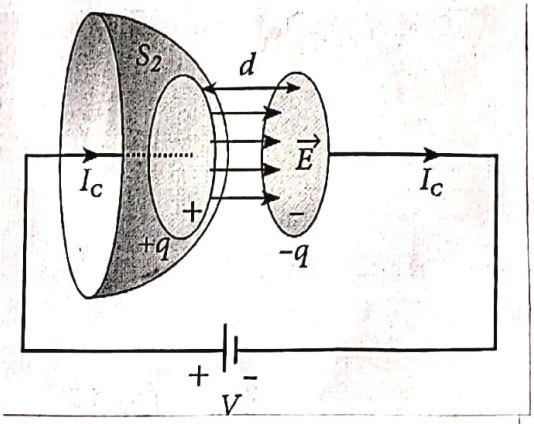


Figure 5.6 Applying Gauss's law between the plates of the capacitor

From (5) and (6) $I_d = \varepsilon_o \frac{d\phi_E}{dt}$ where I_d is known as displacement current.

The displacement current can be defined as the current which comes into play in the region in which the electric field and the electric flux are changing with time.

Maxwell modified Ampere's law as

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 I = \mu_0 (I_c + I_d)$$

Where $I = (I_c + I_d)$ = sum of the conduction current and displacement current.

33. A proton and an electron have same de Broglie wavelength. Which of them moves faster and which possesses more kinetic energy?

The de Broglie wavelength associated with proton = $\lambda_p = \frac{h}{\sqrt{2m_p k_p}}$ .....(1)

The de Broglie wavelength associated with electron = $\lambda_e = \frac{h}{\sqrt{2m_e k_e}}$.....(2)

$$\lambda_p = \lambda_e$$

$$\frac{h}{\sqrt{2m_p k_p}} = \frac{h}{\sqrt{2m_e k_e}}$$

$$m_p k_p = m_e k_e \text{ .....(3)}$$

Kinetic energy of the proton = $k_p = \frac{1}{2} m_p v_p^2$ .....(4)

Kinetic energy of the electron = $k_e = \frac{1}{2} m_e v_e^2$.....(5)

Substitute (4) and (5) in (3)

$$m_p \times \frac{1}{2} m_p v_p^2 = m_e \times \frac{1}{2} m_e v_e^2$$

$$m_p^2 v_p^2 = m_e^2 v_e^2$$

$$\frac{v_p^2}{v_e^2} = \frac{m_e^2}{m_p^2} \; ; \quad \frac{v_p}{v_e} = \frac{m_e}{m_p} \; ;$$

$m_e < m_p$, $v_p < v_e$, electron moves faster than the proton.

PART - IV

Answer all the questions **5 × 5 = 25**

34. (a) What is absorption spectra?. Give their types.

When light is allowed to pass through a medium or an absorbing substance then the spectrum obtained is known as absorption spectrum. It is the characteristic of absorbing substance. Absorption spectrum is classified into three types:

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(i) Continuous absorption spectrum

When the light is passed through a medium, it is dispersed by the prism, we get continuous absorption spectrum. For instance, when we pass white light through a blue glass plate, it absorbs everything except blue. This is an example of continuous absorption spectrum.

(ii) Line absorption spectrum

When light from the incandescent lamp is passed through cold gas (medium), the spectrum obtained through the dispersion due to prism is line absorption spectrum. Similarly, if the light from the carbon arc is made to pass through sodium vapour, a continuous spectrum of carbon arc with two dark lines in the yellow region of sodium vapour is obtained.

(iii) Band absorption spectrum

When the white light is passed through the iodine vapour, dark bands on continuous bright background is obtained. This type of band is also obtained when white light is passed through diluted solution of blood or chlorophyll or through certain solutions of organic and inorganic compounds.

(OR)

b) What are the possible harmful effects of usage of nano particles ? Why?

(i) The major concern here is that the nanoparticles have the dimensions same as that of the biological molecules such as proteins. They may easily get absorbed on to the surface of living organisms and they might enter the tissues and fluids of the body.

(ii) The adsorbing nature depends on the surface of the nanoparticle. Indeed, it is possible to deliver a drug directly to a specific cell in the body by designing the surface of a nanoparticle so that it adsorbs specifically on to the surface of the target cell.

(iii) The interaction with living systems is also affected by the dimensions of the nanoparticles. Nanoparticles of a few nanometers size may reach well inside biomolecules but it is not possible for larger nanoparticles.

(iv) Nanoparticles can also cross cell membranes. It is also possible for the inhaled nanoparticles to reach the blood, to reach other sites such as the liver, heart or blood cells.

(v) Researchers are trying to understand the response of living organisms to the presence of nanoparticles of varying size, shape, chemical composition and surface characteristics.

35. a) i) Explain in detail Coulomb's law and its various aspects.

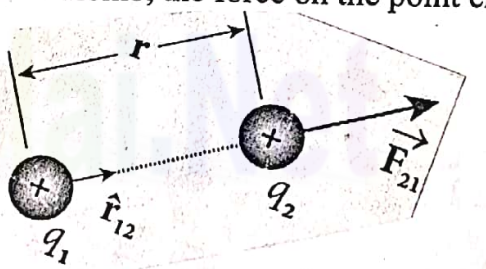
Coulomb deduced the expression for the force between two stationary point charges in vacuum or free space.

Consider two point charges q_1 and q_2 at rest in vacuum, and separated by a distance of r, as shown in Figure .According to Coulomb, the force on the point charge q_2 exerted by another point charge q_1 is

$$\vec{F}_{21} = k\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}$$

where $\hat{r}_{12}$ is the unit vector directed from charge q_1 to charge q_2

k is the proportionality constant.



Important aspects of Coulomb's law

(1) Coulomb's law states that the electrostatic force is directly proportional to the product of the magnitude of the two point charges and is inversely proportional to the square of the distance between the two point charges.

(2) The force on the charge q_2 exerted by the charge q_1 always lies along the line joining the two charges. $\hat{r}_{12}$ is the unit vector pointing from charge q_1 to q_2 .It is shown in the Figure. Likewise, the force on the charge q_1 exerted by q_2 is along $-\hat{r}_{12}$ (i.e., in the direction opposite to $\hat{r}_{12}$).

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(3) Here $k = \frac{1}{4\pi\varepsilon_o} = 9 \times 10^9\ Nm^2\ C^{-2}$

Where is the permittivity of free space or vacuum

$$\varepsilon_o = \frac{1}{4\pi k} = 8.85 \times 10^{-12}\ C^2\ N^{-1}m^{-2}$$

(4) The magnitude of the electrostatic force between two charges each of one

$$|\vec{F}| = \frac{9\times10^9 \times 1\times1}{1^2} = 9 \times 10^9\ N.$$

This is a huge quantity, almost equivalent to the weight of one million ton. We never come across 1 coulomb of charge in practice. Most of the electrical phenomena in day-to-day life involve electrical charges of the order of μC (micro coulomb) or nC (nano coulomb).

(5) In SI units, Coulomb's law in vacuum takes the form

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_o} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

In a medium of permittivity ε, the force between two point charges is given by

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

Since ε > εo, the force between two point charges in a medium other than vacuum is always less than that in vacuum.

We define the relative permittivity for a given medium as $\varepsilon_r = \frac{\varepsilon}{\varepsilon_o}$

For vacuum or air, $\varepsilon_r = 1$ and for all other media $\varepsilon_r > 1$.

(6) Coulomb's law has same structure as Newton's law of gravitation. Both are inversely proportional to the square of the distance between the particles. But there are some important differences between these two laws.

Gravitational force	Coulomb force
1.The gravitational force is directly proportional to the product of two masses.	1. The electrostatic force is directly proportional to the product of the magnitude of two point charges.
2.The gravitational force between two masses is always attractive	2. Coulomb force between two charges can be attractive or repulsive, depending on the nature of charges.
3.The value of the gravitational constant $G = 6.626 \times 10^{-11}\ N\ m^2\ kg^{-2}$.	3. The value of the constant k in Coulomb law is $k = 9 \times 10^9\ N\ m^2\ C^{-2}$. Since k is much more greater than G, the electrostatic force is always greater in magnitude than gravitational force for smaller size objects.
. 4. The gravitational force between two masses is independent of the medium.	4. The electrostatic force between the two charges depends on nature of the medium in which the two charges are kept at rest.
5. The gravitational force between two point masses is the same whether two masses are at rest or in motion.	5. If the charges are in motion, yet another force (Lorentz force) comes into play in addition to coulomb force

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7) Electrostatic force obeys Newton's third law. (i.e)

$$(\vec{F}_{12} = -\vec{F}_{21}$$

ii) Calculate the force between electron and proton in Hydrogen atom($e = 1.6 \times 10^{-19}$ C and $r_o = 0.53$ Å

$$F_e = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r^2} = \frac{9\times10^9\times(1.6\times10^{-19})^2}{(0.53\times10^{-10})^2} = \frac{9\times1.6^2\times10^{-29}}{0.53^2\times10^{-20}} = \frac{9\times2.56\times10^{-9}}{0.2809} = 8.2\times10^{-8}\ \text{N}$$

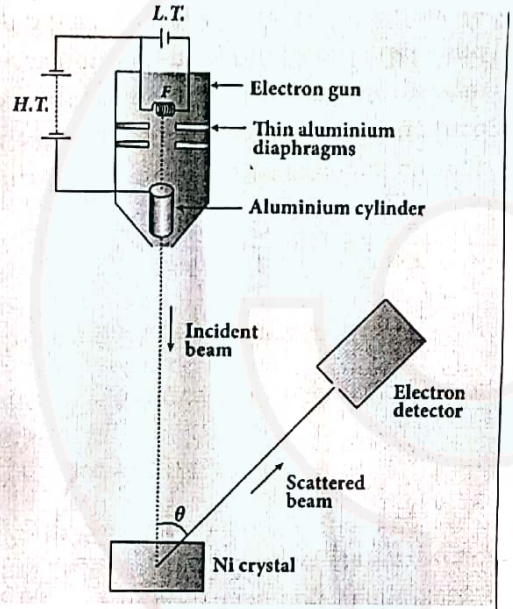
(OR)

b) Describe briefly Davisson – Germer experiment which demonstrated the wave nature of electrons.

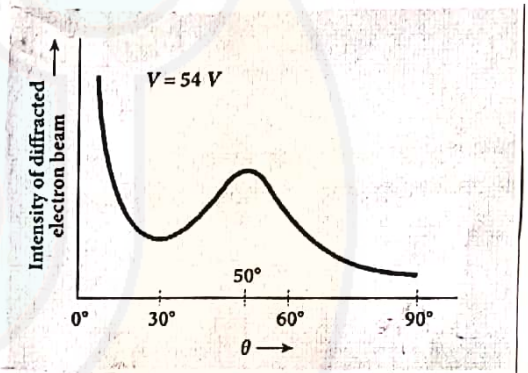
De Broglie hypothesis of matter waves was experimentally confirmed by Clinton Davisson and Lester Germer in 1927.

They demonstrated that electron beams are diffracted when they fall on crystalline solids. Since crystal can act as a three-dimensional diffraction grating for matter waves, the electron waves incident on crystals are diffracted off in certain specific directions.

Figure (1) shows a schematic representation of the apparatus for the experiment.



Experimental set up of Davisson – Germer experiment



Variation of intensity of diffracted electron beam with the angle θ

The filament F is heated by a low tension (L.T.) battery. Electrons are emitted from the hot filament by thermionic emission. They are then accelerated due to the potential difference between the filament and the anode aluminium cylinder by a high tension (H.T.) battery. Electron beam is collimated by using two thin aluminium diaphragms and is allowed to strike a single crystal of Nickel.

The electrons scattered by Ni atoms in different directions are received by the electron detector which measure the intensity of scattered electron beam.The detector is rotatable in the plane of the paper so that the angle θ between the incident beam and the scattered beam can be changed at our will. The intensity of the scattered electron beam is measured as a function of the angle θ.

Figure (2) shows the variation of intensity of the scattered electrons with the angle θ for the accelerating voltage of 54V. For a given accelerating voltage V, the scattered wave shows a peak or maximum at an angle of 50° to the incident electron beam. This peak in intensity is attributed to the constructive interference of electrons diffracted from various atomic layers of the target material. From the known value of inter planar spacing of Nickel, the wavelength of the electron wave has been experimentally calculated as 1.65 Å

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The wavelength can also be calculated from de Broglie relation for V = 54 V from the equation as

$$\lambda = \frac{12.27}{\sqrt{V}}\text{Å} = \frac{12.27}{\sqrt{54}}\text{ Å}$$

$$\lambda = 1.67\text{ Å}$$

This value agrees well with the experimentally observed wavelength of 1.65 Å . Thus this experiment directly verifies de Broglie's hypothesis of the wave nature of moving particles.

36. a) Obtain lens maker's formula and mention its significance

Let us consider a thin lens made up of a medium of refractive index n_2 is placed in a medium of refractive index n_1. Let R_1 and R_2 be the radii of curvature of two spherical surfaces (1) and (2) respectively and P bethe pole as shown in figure.

Consider a point object O on the principal axis. The ray which falls very close to P, after refraction at the surface (1) forms image at I'. Before it does so, it is again refracted by the surface (2).

Therefore the final image is formed at I.

The general equation for the refraction at a spherical surface is

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R} \quad \text{--------(1)}$$

For the refracting surface (1), the light goes from n_1 to n_2.

$$\frac{n_2}{v'} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R_1} \quad \text{----------(2)}$$

For the refracting surface (2) , the light goes from medium n_2 to n_1.

$$\frac{n_1}{v} - \frac{n_2}{v'} = \frac{(n_1 - n_2)}{R_2} \quad \text{----------(3)}$$

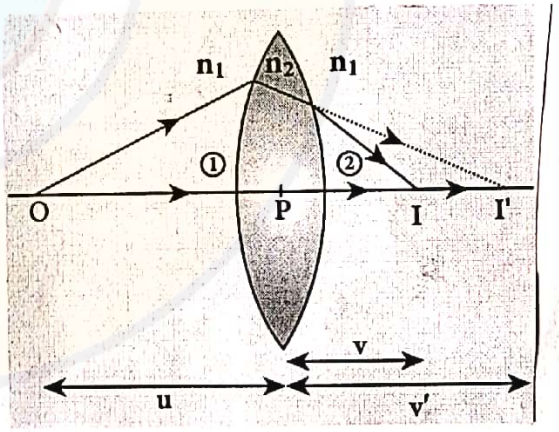
Adding the above two equations (2)and (3)

$$\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{--------(4)}$$

Further simplifying and rearranging,

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2 - n_1}{n_1}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{---------(5)}$$



If the object is at infinity, the image is formed at the focus of the lens. Thus, for $u = \infty$, $v = f$.

Then the equation becomes.

$$\frac{1}{f} - \frac{1}{\infty} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{----(6)}$$

If the refractive index of the lens is n_2 and it is placed in air, then $n_2 = n$ and $n_1 = 1$. So the equation (6.67) becomes,

$$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{----------(7)}$$

The above equation is called the **lens maker's formula**

It tells the lens manufactures what curvature is needed to make a lens of desired focal length with a material of particular refractive index. This formula holds good also for a concave lens.

(OR)

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b) To determine the age of the materials in Keezhadi, the charcoal of 200 g sent for carbon dating. The activity of $^{14}_{6}C$ is found to be 38 decays/s. Calculate the age of charcoal. (Half life of $^{14}_{6}C$ = 5730 years)

$R = R_0 e^{-\lambda t}$; $e^{-\lambda t} = \frac{R}{R_0}$; $e^{\lambda t} = \frac{R_0}{R}$

$\lambda t = \log_e \left(\frac{R_0}{R}\right)$

$t = \frac{1}{\lambda} \log_e \left(\frac{R_0}{R}\right) = \frac{1}{\lambda} \times 2.303 \times \log_{10} \left(\frac{R_0}{R}\right)$----------(1)

$R = 38$ Bq ----------(2)

Initial activity $R_0 = \lambda N_0$----------(3)

$\lambda = \frac{0.6931}{T_{\frac{1}{2}}} = \frac{0.6931}{5730} = 1.21 \times 10^{-4}$ /year----------(4)

$= \frac{0.6931}{5730 \times 365.25 \times 24 \times 60 \times 60} = 3.83 \times 10^{-12}$/s--------(5)

12 gram $^{12}_{6}C$ atom contains = 6.023×10^{23} atoms

1 gram $^{12}_{6}C$ atom contains = 6.023×10^{23} / 12

200 gram $^{12}_{6}C$ atom contains = $\frac{6.023 \times 10^{23}}{12} \times 200 \approx 10^{25}$ ----------(6)

When the sample was alive, the ratio of $^{14}_{6}C$ to $^{12}_{6}C = 1.3 \times 10^{-12} : 1$

Total number of $^{14}_{6}C$ aoms initially = $N_0 = 10^{25} \times 1.3 \times 10^{-12} = 1.3 \times 10^{13} atoms$---(7)

Initial activity = $R_0 = \lambda N_0 = 3.83 \times 10^{-12} \times 1.3 \times 10^{13} \approx 50$ Bq------(8)

From (1) $t = \frac{1}{\lambda} \times 2.303 \times \log_{10} \left(\frac{R_0}{R}\right)$

Substitute (2), (4), (8) in the above equation

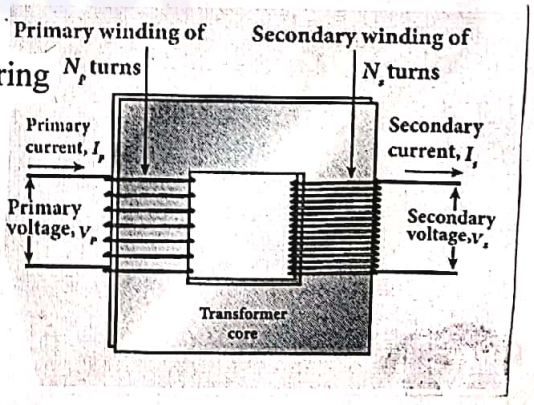
$t = \frac{1}{1.21 \times 10^{-4}} \times 2.303 \times \log_{10} \left(\frac{38}{50}\right)$

$= \frac{2.303 \times 0.1193}{1.21} = 2270$ year

37. a) Explain the construction and working of transformer. Mention the various energy losses in a transformer.

Principle

The principle of transformer is the mutual induction between two coils. That is, when an electric current passing through a coil changes with time, an emf is induced in the neighbouring coil.



Construction

It consists of two coils of high mutual inductance wound over the same transformer core made up of silicone steel.

To avoid eddy current loss, the core is generally laminated

The alternating voltage is applied across primary coil (P), and the output is taken across secondary coil (S)

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The assembled core and coils are kept in a container which is filled with suitable medium for better insulation and cooling purpose.

Working :

The alternating voltage given to the primary coil, set up an alternating magnetic flux in the laminated core.

As the result of flux change, emf is induced in both primary and secondary coils.

The emf induced in the primary coil ε_P is almost equal and opposite to the applied voltage v_p and is given by,

$$v_p = \varepsilon_p = - N_p \frac{d\phi_B}{dt} \text{ ----------(1)}$$

The frequency of alternating magnetic flux is same as the frequency of applied voltage. Therefore induced in secondary will also have same frequency as that of applied voltage.

The emf induced in the secondary coil ε_s is given by

$$\varepsilon_s = - N_s \frac{d\phi_B}{dt} \text{ ----------(2)}$$

where N_P and N_S ,are the number of turns in the primary and secondary coil.

If the secondary circuit is open, then $\varepsilon_s = v_s$ where v_s is the voltage across secondary coil.

$$v_s = \varepsilon_s = - N_s \frac{d\phi_B}{dt} \text{ ----------(3)}$$

From equations (1) and (3), $\frac{v_s}{v_p} = \frac{N_s}{N_p} = K$ .....(4)

This constant K is known as voltage transformation ratio.

For an ideal transformer, Input power = Output power

$$v_p i_p = v_s i_s$$

where i_p and i_s are the currents in the primary and secondary coil respectively.

Therefore, $\frac{v_s}{v_p} = \frac{N_s}{N_p} = \frac{i_p}{i_s}$ ----------(5)

Equation (5) is written in terms of amplitude of corresponding quantities,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = K$$

i) If $N_s > N_p$ (or $K > 1$), $\therefore V_s > V_p$ and $I_s < I_p$. This is the case of step-up transformer in which voltage is increased and the corresponding current is decreased.

ii) If $N_s < N_p$ (or $K < 1$), $\therefore V_s < V_p$ and $I_s > I_p$ .This is step-down transformer where voltage is decreased and the current is increased.

Energy losses in a transformer:

i) Core loss or Iron loss
ii) Copper loss
iii) Flux leakage

(OR)

b) Using Biot Savert law, obtain a relation for the magnetic induction at a point along the axis of a circular coil carrying current .

Consider a current carrying circular loop of radius R and let I be the current flowing through the wire in the direction as shown in Figure. The magnetic field at a point P on the axis of the circular coil at a distance *z* from its center of the coil O. It is computed by taking two diametrically opposite line elements of the coil each of length at C and D. Let be the vector joining the current element (I) at C to the point P.

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$PC = PD = r = \sqrt{R^2 + Z^2}$

$\angle CPO = \angle DPO = (90 - \theta)$

According to Biot-Savart's law, the magnetic field at P due to the current element $Id\vec{\ell}$ is

$$d\vec{B} = \frac{\mu_o}{4\pi} \frac{I\, d\vec{\ell} \times \hat{r}}{r^2}$$

The magnitude of magnetic field due to current element $Id\vec{\ell}$ at C and D are equal because of equal distance from the coil.

The magnetic field $d\vec{B}$ due to each current element $Id\vec{\ell}$ is resolved into two components; dB sin θ along y - direction and dB cos θ along z - direction. Horizontal components of each current element cancels out while the vertical components (dB cos θ $\hat{k}$) alone contribute to total magnetic field at the point P.

If we integrate around the loop, $d\vec{B}$ sweeps out a cone as shown in Figure, then the net magnetic field at point P is

$$\vec{B} = \int d\vec{B} = \int dB \cos\theta\, \hat{k}$$

$$\vec{B} = \frac{\mu_o I}{4\pi} \int \frac{d\ell}{r^2} \cos\theta\, \hat{k}$$

$$= \frac{\mu_o I}{4\pi} \frac{1}{r^2} \cos\theta\, \hat{k} \int d\vec{\ell}$$

using Pythagorous theorem

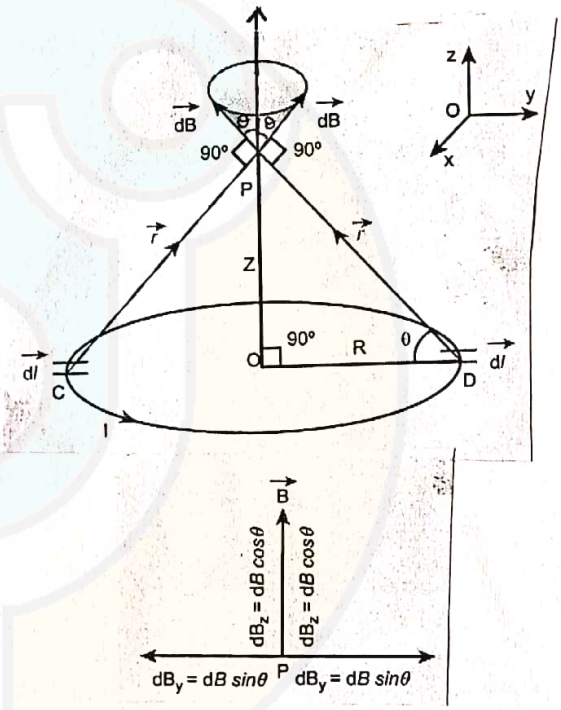
$$r^2 = R^2 + Z^2, \quad \cos\theta = \frac{R}{(R^2+Z^2)^{\frac{1}{2}}}$$

and integrating line element from 0 to 2πR,

$$\vec{B} = \frac{\mu_o I}{4\pi} \frac{1}{r^2} \cos\theta\, \hat{k} \int_0^{2\pi R} d\vec{\ell}$$

$$= \frac{\mu_o I}{4\pi} \frac{1}{(R^2+Z^2)} \frac{R}{(R^2+Z^2)^{\frac{1}{2}}} \hat{k}\,(2\pi R)$$

we get $$\vec{B} = \frac{\mu_o I}{2} \frac{R^2}{(R^2+Z^2)^{\frac{3}{2}}} \hat{k}$$



Note that the magnetic field points along the direction from the point O to P.

Suppose if the current flows in clockwise direction, then magnetic field points in the direction from the point P to O.

38. a) Obtain the condition for bridge balance in Wheatstone's bridge.

An important application of Kirchhoff 's rules is the Wheatstone's bridge.

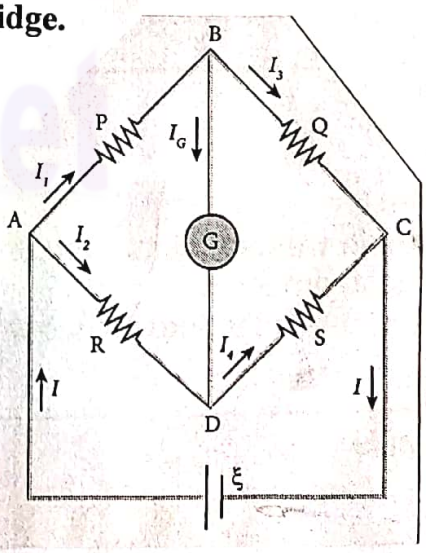
It is used to compare resistances and also helps in determining the unknown resistance in electrical network.

The bridge consists of four resistances P, Q, R and S connected as shown in Figure .

A galvanometer G is connected between the points B and D.

The battery is connected between the points A and C.

The current through the galvanometer is I_G and its resistance is G.



Applying Kirchhoff 's current rule to junction B

$$I_1 - I_G - I_3 = 0 \ldots\ldots\ldots\ldots (1)$$

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Applying Kirchhoff 's current rule to junction D,

$I_2 + I_G - I_4 = 0$..... (2)

Applying Kirchhoff 's voltage rule to loop ABDA,

$I_1P + I_GG - I_2R = 0$ .....(3)

Applying Kirchhoff 's voltage rule to loop ABCDA,

$I_1P + I_3Q - I_4S - I_2R = 0$ .....(4)

When the points B and D are at the same potential, the bridge is said to be balanced. As there is no potential difference between B and D, no current flows through galvanometer ($I_G = 0$). Substituting $I_G = 0$ in equation (1), (2) and (3), we get

$I_1 = I_3$ .....(5)

$I_2 = I_4$ .....(6)

$I_1P = I_2R$ .....(7)

Substituting the equation (5) and (6) in equation (7)

$I_1P + I_1Q - I_2S - I_2R = 0$

$I_1 (P + Q) = I_2 (R + S)$ ---------(8)

Dividing equation (8) by equation (7), we get

$$\frac{I_1(P+Q)}{I_1P} = \frac{I_2(R+S)}{I_2R}$$

$$\frac{P+Q}{P} = \frac{R+S}{R}$$

$$1 + \frac{Q}{P} = 1 + \frac{S}{R}$$

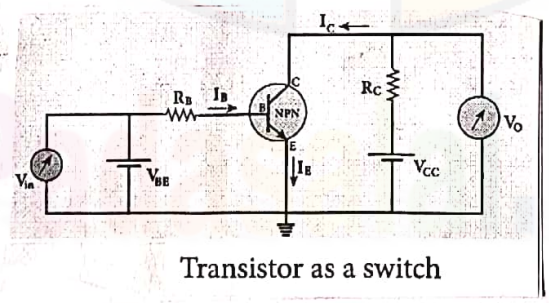
$$\frac{Q}{P} = \frac{S}{R} \quad \text{(or)} \quad \frac{P}{Q} = \frac{R}{S}$$

This is the bridge balance condition.

(OR)

b) i) Transister functions as a switch. Explain Transistor as a switch

The transistor in saturation and cut-off regions functions like an electronic switch that helps to turn ON or OFF a given circuit by a small control signal.



Transistor as a switch

Presence of dc source at the input (saturation region):

When a high input voltage (V_{in} = +5V) is applied, the base current (I_B) increases and in turn increases the collector current.

The transistor will move into the saturation region (turned ON).

The increase in collector current (I_C) increases the voltage drop across R_C , thereby lowering the output voltage, close to zero.

The transistor acts like a closed switch and is equivalent to ON condition.

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Absence of dc source at the input (cutoff region):

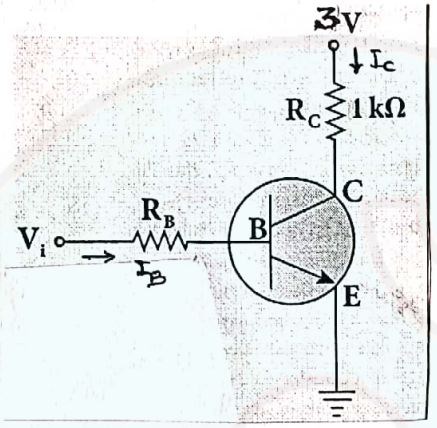
A low input voltage (V_{in} =0 V), decreases the base current (I_B) and in turn decreases the collector current (I_C).

The transistor will move into the cut-off region (turned OFF).

The decrease in collector current (I_C) decreases the drop across R_C , thereby increasing the output voltage, close to +5 V.

The transistor acts as an open switch which is considered as the OFF condition.

ii) Assuming V_{CEsat}= 0.2 V and β = 50 , find the minimum base current (I_B) required to derive the transistor given in the figure to saturation



V_{cc} = 3V ; R_c = 1kΩ = 1000 Ω

$V_{CC} = I_c R_c + V_{CC}$

$$I_C = \frac{V_{CC}-V_{CE}}{R_C} = \frac{3-0.2}{1000} = 2.8\times 10^{-3}\ A$$

$$\beta = \frac{I_C}{I_B}\ ;\ I_B = \frac{I_C}{\beta} = \frac{2\times 2.8\times 10^{-3}}{100} = 5.6\times 10^{-5}\ A = 56\times 10^{-6}\ A = 56\mu A$$

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